Curtis Ying

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Education

Brown University — **Expected Graduation May 2028**

*B.Sc. Applied Mathematics and Computer Science (**GPA: 4.00/4.00**)*

Organizations: Head TA (Databases), TA (Computer Systems), Full Stack Club (web dev), Hack@Brown, Forecasting Club

Experience

Software Engineer Intern — May - August 2026

Pinterest

- Shipped CI testing and build optimizations for Pinterest iOS platform using caching, selective testing, and dynamic sharding, **decreasing build and test time by 40% and infrastructure load by 10%**
- Migrated testing pipelines to Tuist open-source toolkit and collaborated closely with Tuist developers to resolve bugs and scope new features
- **Managed resolution of SEV-3 incident** causing all iOS CI builds to fail, determining root cause in Tuist repository, deploying **stable fix to pipelines within 90 minutes**, and leading post-mortem analysis meeting

Lead AI Undergraduate Researcher — June 2025 – Present

Brown University Lee Lab

- Led and collaborated with a small team researching **multimodal ML-LLM model** to predict operating room (OR) usage time and optimize scheduling, with planned **deployment to Rhode Island Hospital**
- Developed pipelines using Langchain, Transformers, and PyTorch with **AutoEncoder/RAG** and **RL fine-tuning** for processing of tabular and textual clinical data
- Combined processed text with structured clinical data into downstream FNN to create fully trainable end-to-end model

AI and Software Engineer Intern — May – August 2025

New Horizon Soft, LLC

- Created root-cause analysis algorithm for inventory shortages and surpluses to optimize stock levels, **reducing time to investigate inventory issues by 60% for Chick-fil-A**
- Designed and built customized AI agent for clients through iterative sprints, Python LangChain, SQL, and **MCP framework** to retrieve relevant data from Oracle databases with LLMs

Projects

AI Google Review Agent — **Completed June 2025 | Github**

- Built a Chrome extension featuring a custom AI chatbot and summarizer for Google Maps locations and reviews
- Utilized Qdrant database, Cohere, and **RAG** to retrieve reviews relevant to user queries and generate responses using LLMs, with **Flask server** connecting to Google review database and JavaScript frontend for Chrome Extension UI
- **Decreased search time** on average by over **75%** for summary-based queries and **90%** for custom/complex queries

Website for Brown Opinion Project — **Completed April 2025 | GitHub | Demo**

- Frontend developer on teams, polls, and news pages, in collaboration with small team at Full Stack at Brown
- Built animations, mobile-friendly UI components like dropdowns and collapsible sections using **JS**, **CSS**, and **Tailwind**

Publications

Ying, C., & Liebson, M. (2024). An innovative approach to optimize inbound delivery scheduling at distribution centers. *International Journal of Operations Research and Information Systems*, *15*(1), 1-22. https://doi.org/10.4018/ijoris.345920

Recognitions

Brown Math Modeling Competition (BMCM) — **November 2024 – January 2025**

- Outstanding Winner out of 22 collegiate teams from Brown, Duke, UNC, OSU, and NC State

MathWorks Math Modeling Challenge (M3C) — **March 2023 – April 2024**

- 2x Semifinalist (Top 2% of 650+ teams), Technical Computing 2nd Place & Honorable Mention

Skills

Languages: Python, C/C++, MATLAB, Java, Go, HTML/CSS, JavaScript, SQL, Swift/XCode, Pyret

Frameworks/Tools: MCP, PyTorch, TensorFlow, React, Pandas, Full Stack, Flask, Docker, Next.js, Node.js, NumPy, Sklearn, Git, OracleSQL, HuggingFace, Assembly (x86), GDB, CUDA, Tuist, BuildKite